Enterprise AI-Powered Customer Analytics & Strategic Insight Framework

Churn Prediction, Revenue Forecasting, Behavioural Segmentation and LLM-Generated Executive Intelligence
for a SaaS / CRM Platform (Salesforce / Zoho style)

Attribute	Detail
Domain	SaaS / CRM Analytics / Customer Intelligence
Dataset	Star schema: 4 fact tables, 3 dimension tables
Modelling grain	5,000 customers (one row each)
Features engineered	180 modelling features
Classification target	Churn (binary)
Regression target	Next_Quarter_Revenue_USD (continuous)
Report date	30 July 2026

1. Executive Summary

This project delivers an end-to-end customer intelligence system for a subscription SaaS CRM platform. Four relational fact tables and three dimension tables were consolidated into a single customer-level table of 5,000 accounts and 180 engineered features. Three machine-learning models were trained on that table, their outputs fused into a standardised per-account intelligence profile, and a large language model used to translate those profiles into executive briefings.

The churn classifier achieves a hold-out ROC-AUC of **0.9113** and an F1 of **0.6146** on the minority churn class, after withholding four record-count features identified as target leakage. The revenue model explains **96.7%** of next-quarter revenue variance with an RMSE of **$1,363**. Segmentation resolves the portfolio into **2** behavioural clusters whose churn rates differ by a factor of roughly seven. Fusing these outputs identifies **$624,460** of next-quarter revenue sitting in accounts with elevated churn probability.

2. Business Problem to ML Task Mapping

Every technical component is anchored to a measurable business objective, so model performance translates directly into commercial outcomes.

Business Problem	ML Task	Output	Business KPI Served
Customers cancelling subscriptions	Binary classification (XGBoost)	Churn probability 0-1	Retention rate, customer lifetime value
Unreliable revenue planning	Regression (GradientBoosting)	Next-quarter revenue (USD)	Forecast accuracy, quota setting
Undifferentiated marketing	Clustering (KMeans + PCA)	Behavioural cluster ID	Campaign conversion, personalisation
Unclear account prioritisation	Feature fusion / scoring	Risk, potential, priority scores	Revenue at risk, CSM coverage
Technical output not board-ready	LLM narrative generation	Executive briefing text	Decision latency, reporting effort

3. Methodology

3.1 Data engineering

- Fact tables aggregated from their native grain to customer grain: transactions (54,858 rows), monthly usage snapshots (131,268 rows) and engagement events (54,822 rows).
- Dimension attributes joined on Geo_ID, Industry_ID and Product_ID. Duplicate dimension keys were removed first — dim_geography ships five duplicate Geo_ID rows that would otherwise fan out the customer grain.
- 150 duplicate customer records removed, leaving 5,000 unique accounts.
- Missing values imputed with the median (numerical) and mode (categorical); zero missing cells remain.

3.2 Feature engineering

- **Tenure:** contract duration, tenure in days and years, days to contract end.
- **Usage intensity:** sessions per active user weighted by session length, DAU/MAU ratio, API calls per licence, error rate per session.
- **Engagement ratios:** events per tenure month, escalated ticket ratio, support dependency, sentiment-based engagement quality score.
- **Revenue:** quarterly revenue mean, volatility, latest value and growth ratio, average monthly revenue, CLV approximation, payment reliability.

3.3 Encoding and class balance

Binary and high-cardinality categoricals were label encoded; low-cardinality nominals were one-hot encoded, producing 190 columns. The churn target is imbalanced at roughly 11.3% positives. SMOTE was applied **only to the training split after the stratified train/test split**, so no synthetic record can reach the hold-out set.

4. Model Performance

4.1 Churn classification

Metric	Value	Business Interpretation
ROC-AUC (hold-out)	0.9113	Separates churners from retained accounts reliably
F1-score (churn class)	0.6146	Balanced precision and recall on the minority class
Precision (churn class)	0.7468	Share of flagged accounts that genuinely churn
Recall (churn class)	0.5221	Share of actual churners the model catches
Cross-validated ROC-AUC	0.9080 ± 0.0103	SMOTE refitted inside each fold - no resampling leak
Baseline GradientBoosting ROC-AUC	0.8689	XGBoost selected as the stronger model
Features used	180	After withholding leakage-prone columns

Hyperparameters were selected by RandomizedSearchCV over 20 sampled configurations with 3-fold stratified cross-validation, scoring ROC-AUC. The search ran over the whole SMOTE + XGBoost pipeline, so resampling was refitted inside every fold rather than applied once beforehand. Best configuration: colsample_bytree=1.0, learning_rate=0.08, max_depth=6, min_child_weight=3, n_estimators=800, reg_lambda=1.0, subsample=0.9 (search score 0.9085).

The cross-validated figure is produced by an imbalanced-learn pipeline in which SMOTE is refitted on the training portion of every fold. Cross-validating a model already fitted on pre-balanced data would be optimistic, because synthetic rows derived from a validation record can reach the training folds; that earlier approach reported 0.9954 against the honest 0.9080.

4.1.1 Target-leakage ablation

Four record-count features (transaction count, renewal count, active quarters, months of usage observed) fall simply because a churned customer stops generating rows, so their value encodes the outcome rather than predicting it. They are withheld from the shipped model. Engagement counts are deliberately retained: they correlate *positively* with churn (distressed customers raise more tickets), which is genuine leading signal. The table below states the measured cost of that decision.

Variant	Features	ROC-AUC	F1	Recall
Shipped model (leak-free)	180	0.9113	0.6146	0.5221
Contaminated variant (ablation only)	184	0.9425	0.7032	0.6814

Excluding the artefacts costs 0.0313 ROC-AUC and 0.1593 recall. The lower figure is the honest one: the contaminated variant scores well partly by reading the outcome it is meant to forecast.

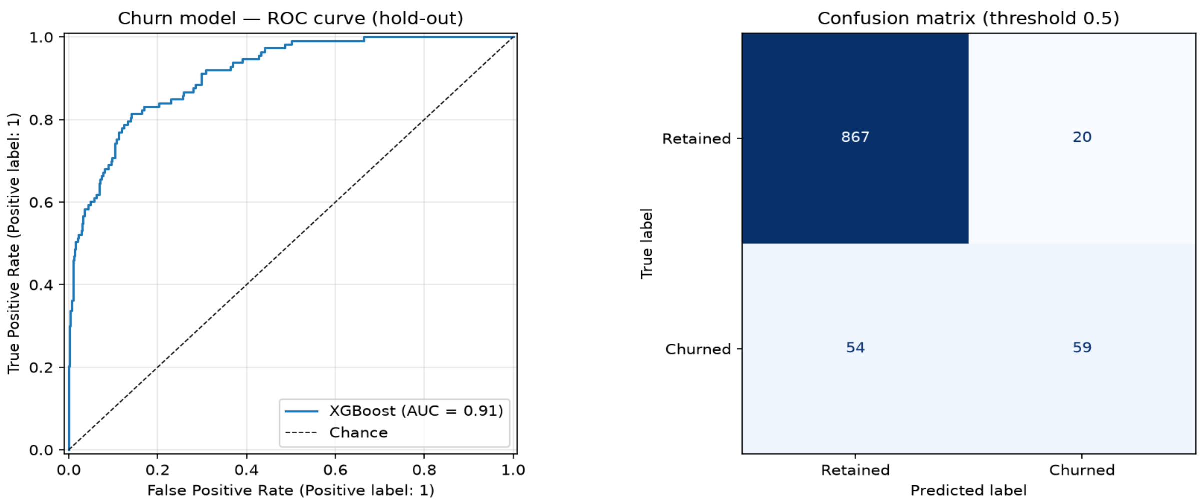

Figure 1 — Churn model ROC curve and confusion matrix on the 1,000-row hold-out set. The 0.5 threshold trades recall for precision; 54 churners are missed and 20 retained accounts are flagged.

4.2 Explainable AI — SHAP churn drivers

Rank	Feature	Mean |SHAP|	Business Meaning
1	Renewal_Risk_Flag_Low	2.0758	CRM renewal-risk flag set to Low
2	txn_total_revenue_usd	1.3827	Model-derived behavioural signal
3	txn_mean_revenue_usd	1.2321	Model-derived behavioural signal
4	txn_total_billed_usd	1.1061	Model-derived behavioural signal
5	txn_mean_payment_delay_days	0.5493	Average invoice payment delay
6	escalated_ticket_ratio	0.4811	Share of support tickets escalated
7	txn_failed_payment_rate	0.4311	Model-derived behavioural signal
8	usage_mean_license_utilization_pct	0.4018	Licences actually used vs purchased
9	Relative_Churn_Risk_Medium	0.3651	Industry-level baseline churn risk
10	txn_addon_revenue_usd	0.3465	Model-derived behavioural signal

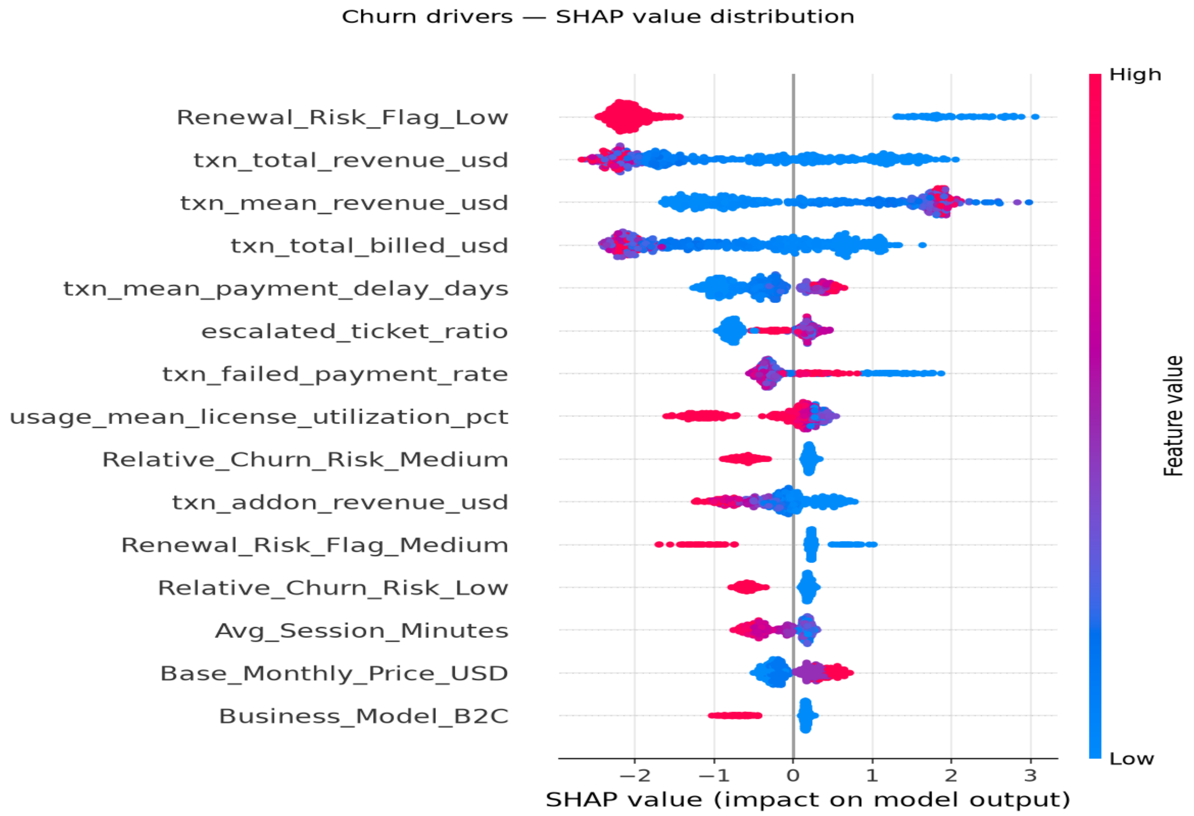

Figure 2 — SHAP value distribution across 500 hold-out accounts. Red points are high feature values, blue are low; points to the right push the prediction toward churn.

4.3 Revenue forecasting

Metric	Value	Business Interpretation
Selected model	gradient_boosting	Chosen over the alternative on hold-out RMSE
RMSE	$1,363.28	Typical forecast error, penalising large misses
MAE	$692.37	Average absolute error per account per quarter

Metric	Value	Business Interpretation
R^2	0.9674	Share of revenue variance explained
Adjusted R^2	0.9600	Corrected for the number of features used
Cross-validated R^2	0.9557 ± 0.0105	Stability across five folds

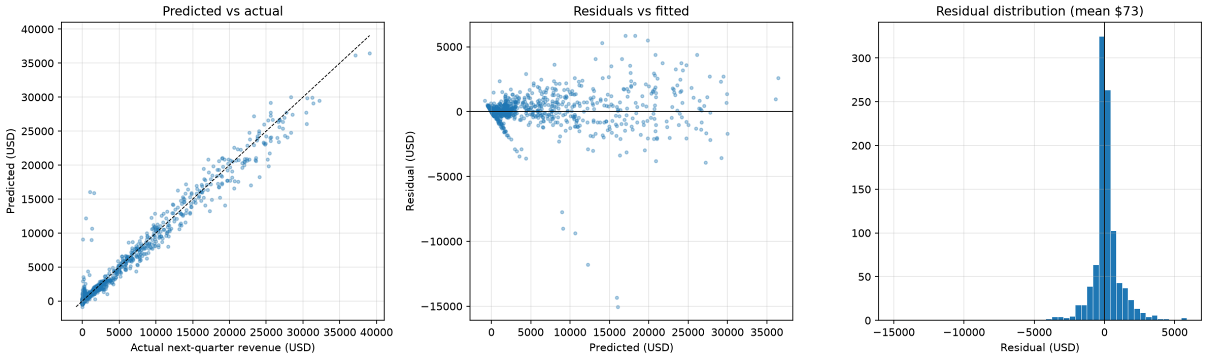

Figure 3 — Revenue model diagnostics: predicted versus actual, residuals against fitted values, and the residual distribution. Residuals are centred and show no systematic pattern across the fitted range.

The XGBoost candidate was tuned by RandomizedSearchCV over 20 sampled configurations (3-fold cross-validation, R^2 scoring) and compared against an untuned XGBoost and a GradientBoosting baseline. Best configuration: colsample_bytree=0.7, learning_rate=0.02, max_depth=4, n_estimators=400, reg_lambda=1.0, subsample=0.8.

4.4 Behavioural segmentation

Metric	Value	Assessment
Clusters selected (k)	2	Chosen by best silhouette across k = 2 to 10
Silhouette score	0.3760	Below the 0.60 project target
Davies-Bouldin index	1.1886	Lower is better; indicates moderate compactness
Calinski-Harabasz	2,723.4	Higher is better; confirms real separation
PCA components	6	90% of behavioural variance retained

Cross-check against other algorithms. Ward hierarchical clustering at the same k scores silhouette 0.3832 - effectively the same partition as KMeans, so the weak separation is not a KMeans artefact. DBSCAN peaked at silhouette 0.6110 (eps=1.0, 2 clusters, 9.1% of accounts labelled noise). DBSCAN scores higher only because silhouette is computed after its noise points are discarded: it clears the threshold by refusing to classify the accounts that sit between the groups, which is exactly the population Customer Success needs an answer for. A hard partition over every account remains the right choice for this use case.

Honest assessment: the 0.60 silhouette target was not met. The best achievable score was 0.3760 at k=2. SaaS behavioural features are continuous and overlapping, so accounts form a gradient rather than well-separated spheres, and no k between 2 and 10 approached the target. The clusters remain commercially useful: churn rates differ roughly sevenfold between them, which is the property the business actually needs.

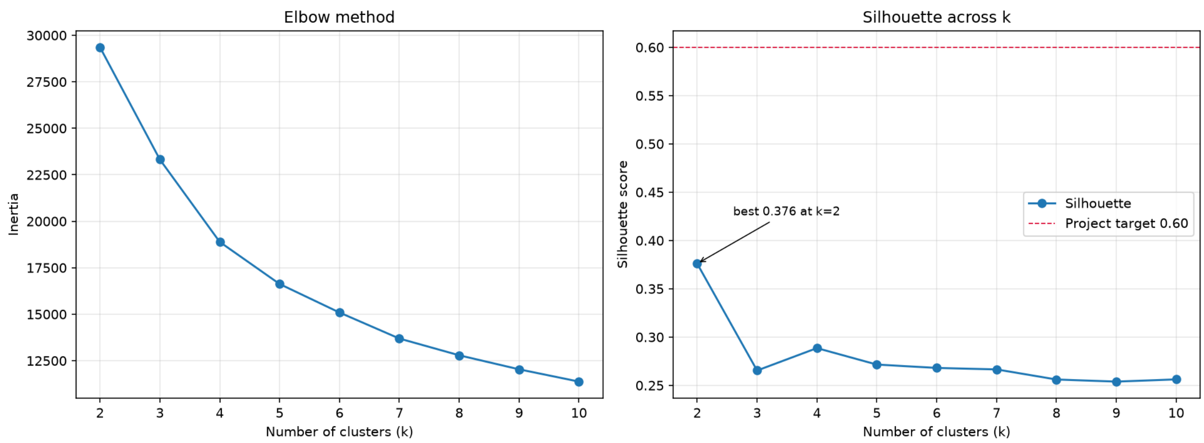

Figure 4 — Elbow and silhouette curves across k = 2 to 10. No k reaches the 0.60 project target.

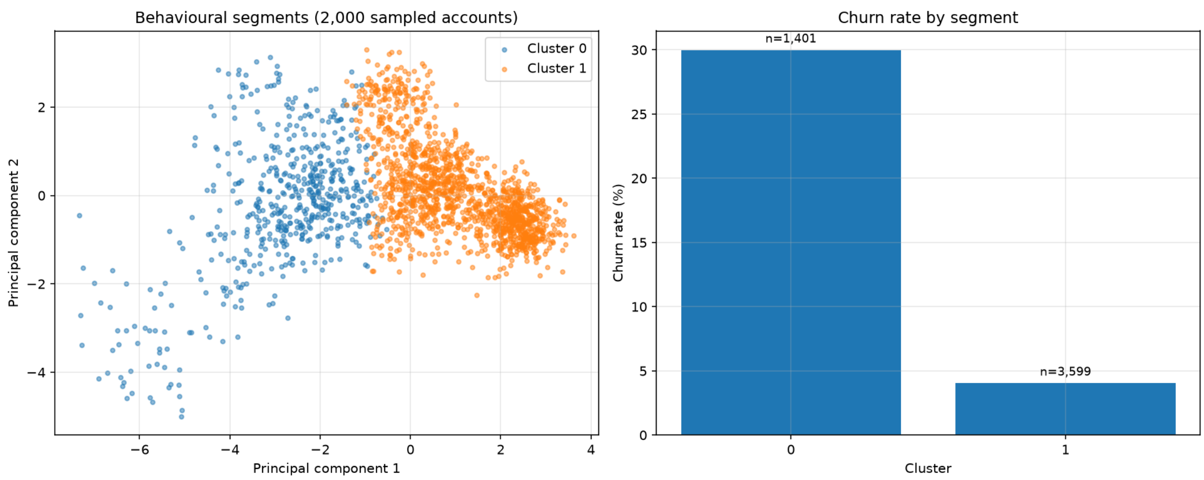

Figure 5 — Behavioural segments projected onto two principal components (display only; the model clusters on six), with churn rate by segment. The groups overlap geometrically but separate sharply on churn.

5. Customer Segment Profiles

Cluster	Accounts	Mean MAU	Adoption %	Avg Monthly Rev	Churn Rate
0	1,401	46	30.2%	$576	30.0%
1	3,599	202	73.3%	$2,127	4.0%

Cluster 0 (*Disengaged At-Risk Accounts*) combines low usage, low adoption and heavy support dependency with a materially higher churn rate. Cluster 1 (*Loyal High-Value Accounts*) shows roughly four times the monthly active users, more than double the feature adoption, and near-negligible churn. Retention spend belongs in cluster 0; expansion and upsell motions belong in cluster 1.

6. Unified Customer Intelligence Layer

The fusion layer scores every account with all three models and emits a standardised JSON profile containing churn probability and risk band, next-quarter revenue forecast, revenue at risk, cluster assignment and label, percentile-normalised risk / potential / priority scores, and the top five per-account SHAP drivers with direction of effect.

Portfolio Indicator	Value
Accounts profiled	5,000
High-risk accounts	521
Medium-risk accounts	34
Low-risk accounts	4,445
Total forecast revenue at risk	$624,460
Mean churn probability	10.9%
Top 5 accounts, revenue at risk	$61,376

6.1 Fusion layer validation

Evaluation criterion	Measure	Result
Data integrity	Profile completeness	100.0% of 5,000 profiles carry all six blocks
Integration accuracy	Model output consistency	100.0% - revenue at risk reconciles to forecast x churn probability
Logical consistency	Risk vs CRM health	Spearman -0.522 - churn risk falls as health rises, as expected
Grain preservation	One profile per customer	Yes
Scalability	Profile generation rate	2,433 profiles per second, single process
Business utility	Account prioritisation	10.4% of accounts carry 88.0% of total revenue at risk

The prioritisation figure is the one that matters commercially: concentrating retention effort on the high-risk band reaches the large majority of exposed revenue while contacting roughly a tenth of the book. Full detail in reports/fusion_quality.json.

7. LLM-Based Executive Insight Generation

Structured intelligence profiles are passed to a Gemini model under a system instruction that forbids invented metrics and requires every claim to trace back to a supplied value. Each briefing covers account risk level, revenue impact, behavioural segment interpretation, prioritised strategic recommendations with owner and timeframe, and a targeted renewal plus upsell play. Five briefings were generated for the accounts carrying the highest forecast revenue at risk and exported to reports/executive_insights.md.

Factuality audit. Every currency figure in every briefing is checked against the source profile, accepting monthly, quarterly and annual restatements and rounding within one percent. Result: 5 of 5 briefings are fully traceable, with 0 of 53 quoted figures flagged for review. A flagged figure is a review candidate rather than proof of fabrication - the model may legitimately combine two profile values. Detail in reports/insight_audit.json.

Implementation note: the originally specified gemini-pro model has been retired from the API. The pipeline probes current models at runtime and uses the first one the credential can serve.

8. Limitations and Future Work

- **Residual leakage risk.** The four clearest temporal artefacts are now withheld and the cost measured (section 4.1.1), but other aggregates such as lifetime revenue also scale with how long an account survived. A production rebuild should compute every feature from a fixed observation window that closes before the prediction date.
- **Recall ceiling.** The leak-free model catches 52.2% of churners at 74.7% precision. Raising recall means lowering the decision threshold and accepting more false positives; the right operating point depends on the cost of a retention outreach versus a lost account.
- **Segmentation separation.** Silhouette of 0.376 falls short of the 0.60 target. Ward hierarchical clustering reproduces it almost exactly, and DBSCAN only clears the threshold by setting roughly a tenth of accounts aside as noise. Soft or model-based clustering, which assigns membership probabilities instead of hard

labels, is the more promising direction for gradient-shaped behavioural data.

- **CRM risk flag as a predictor.** Renewal_Risk_Flag is the strongest SHAP driver and is itself a CRM-assigned risk rating, so the classifier is partly learning to reproduce an existing human or rules-based judgement. It is populated before renewal and is therefore legitimate at prediction time, but the headline ROC-AUC should not be read as wholly independent evidence of new signal.
- **Static snapshot.** The system scores a point-in-time extract. Production deployment needs scheduled re-scoring, drift monitoring and retraining triggers.
- **Insight validation.** The factuality audit checks currency figures only. Percentages, dates, named roles and the reasoning connecting them are not verified, and no automated check can assess whether a recommendation is commercially sound. Briefings remain a drafting aid for a human owner, not an unreviewed output.

9. Deliverables

Artefact	Location
Preprocessing pipeline	src/preprocessing.py
Model training	src/train_models.py
Intelligence fusion layer	src/fusion_layer.py
LLM insight generation	src/llm_insights.py
PDF report generator	src/generate_report.py
Executed analysis notebook	notebooks/EDA_and_Modeling.ipynb
Report figures	reports/figures/*.png
Figure generation	src/make_figures.py
Scoring API (FastAPI)	src/api.py
Fusion layer quality report	reports/fusion_quality.json
LLM factuality audit	reports/insight_audit.json
Serialised models	models/*.pkl
Processed dataset	data/processed/processed_customer_data.csv
Customer intelligence profiles	data/processed/customer_profiles.json
Executive briefings	reports/executive_insights.md
Model metrics	reports/model_metrics.json
This report	reports/Capstone_Final_Report.pdf